www.aktunotes.in

Printed pages: 2 **Sub Code: NEC-302**

Paper Id: | 3 | 0 | 4 | 5 | **Roll No.** | | | | | | | | | | |

B.Tech.
(SEM III) THEORY EXAMINATION 2017-18
Fundamental of Electronic Devices

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.
2. Any special paper specific instruction.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a) Define the semiconductors on the basis of energy band gap with neat sketch?
b) In a simple cubic lattice find the ratio of intercepts on the three axes by (1, 3, 2) Plane?
c) The Minority carrier life time in p type material is 10^{-7}s. The mobility of electron in silicon is 0.15 $m^2V^2s^{-1}$ at 300K. What is the diffusion length?
d) What type of semiconductor material is suitable for luminescence effect?
e) Why is collector current slightly less than emitter current?
f) A JFET has drain current of 5 mA. If I_{DSS} =10 mA and $V_{GS\ off}$ = -6V find the value of V_{GS} and V_P?
g) Why is it not possible to construct a germanium P-N-P-N switching diode?
h) Why SCR can't be used as a bidirectional switch?
i) Define punch-off voltage?
j) Define Quasi Fermi level?

SECTION B

www.aktunotes.in

2. Attempt any *three* of the following: 10 x 3 = 30

a) In an N-type silicon sample, the Fermi Level is 0.3 eV below the conduction band edge. Find the electron and hole concentration in the sample at room temperature. For Si, E_g=1.1 eV $n_i = 1.5x10^{10}\ cm^{-3}$

b) Describe the solar cell? Why must a solar cell be operated in the fourth quadrant of the junction V-I characteristic?

c) What is steady state carrier generation? Prove that $\delta n = g_{op}\tau_n$ where g_{op} is optical generation rate and τ_n is the life time of carrier?

d) What is meant by unit cell and lattice constant? Show that for a cubic lattice, the lattice constant a is given by $a = \left(\frac{nM}{\rho N}\right)^{1/3}$ symbols has usual meaning.

e) Sketch and explain the construction and operation of a MISFET? Discuss its characteristic?

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

a) (i) Discuss diffusion and drift of carriers?

www.aktunotes.in

www.aktunotes.in

(ii)Derive the Einstein's relation?

b) Prove that the concentration of free electrons in an intrinsic semiconductor is given by $n=N_c e^{-(EC-EF)/kT}$.

4. Attempt any *one* part of the following: 10 x 1 = 10

a) What is meant by carrier lifetime? Derive the continuity equation?

b) (i)What is the Hall Effect and its application?

(ii) Derive the relation $\theta_H = \tan^{-1}\theta(\mu B)$?

5. Attempt any *one* part of the following: 10 x 1 = 10

a) What do you mean by forward and reverse biased junction? Give a qualitative description of current flow at the junction? In a PN Junction, the reverse biased current is very small but it is increases abruptly beyond a certain reverse bias, Explain?

b**) (i)** Draw the circuit diagram of a CE amplifier in BJT? Explain the mechanism of amplification?

(ii) Distinguish between Zener breakdown and Avalanche breakdown?

6. Attempt any *one* part of the following: 10 x 1 = 10

a) What are the different modes of operation in MOSFET? Draw the static drain characteristic and transfer characteristic curves for N-channel enhancement type MOSFET?

b) Explain the difference- (i) MOSFET and IGBT (ii) TRIAC and SCR

7. Attempt any *one* part of the following: 10 x 1 = 10

a) Explain the V-I characteristic of a photodiode? Why is GaAs an important semiconductor for the fabrication of the Gunn diode?

www.aktunotes.in

b) (i) Compare Zener diode and tunnel diode?

(ii) Explain the Degenerate Semiconductor?

www.aktunotes.in